

Technical Document #533: Software Engineering - Comprehensive Overview

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Refactoring improves code structure without changing behavior. It makes code easier to understand, maintain, and extend.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Refactoring improves code structure without changing behavior.
- Continuous integration (CI) automatically builds and tests code changes.
- Technical debt represents the cost of choosing easy solutions over better ones.